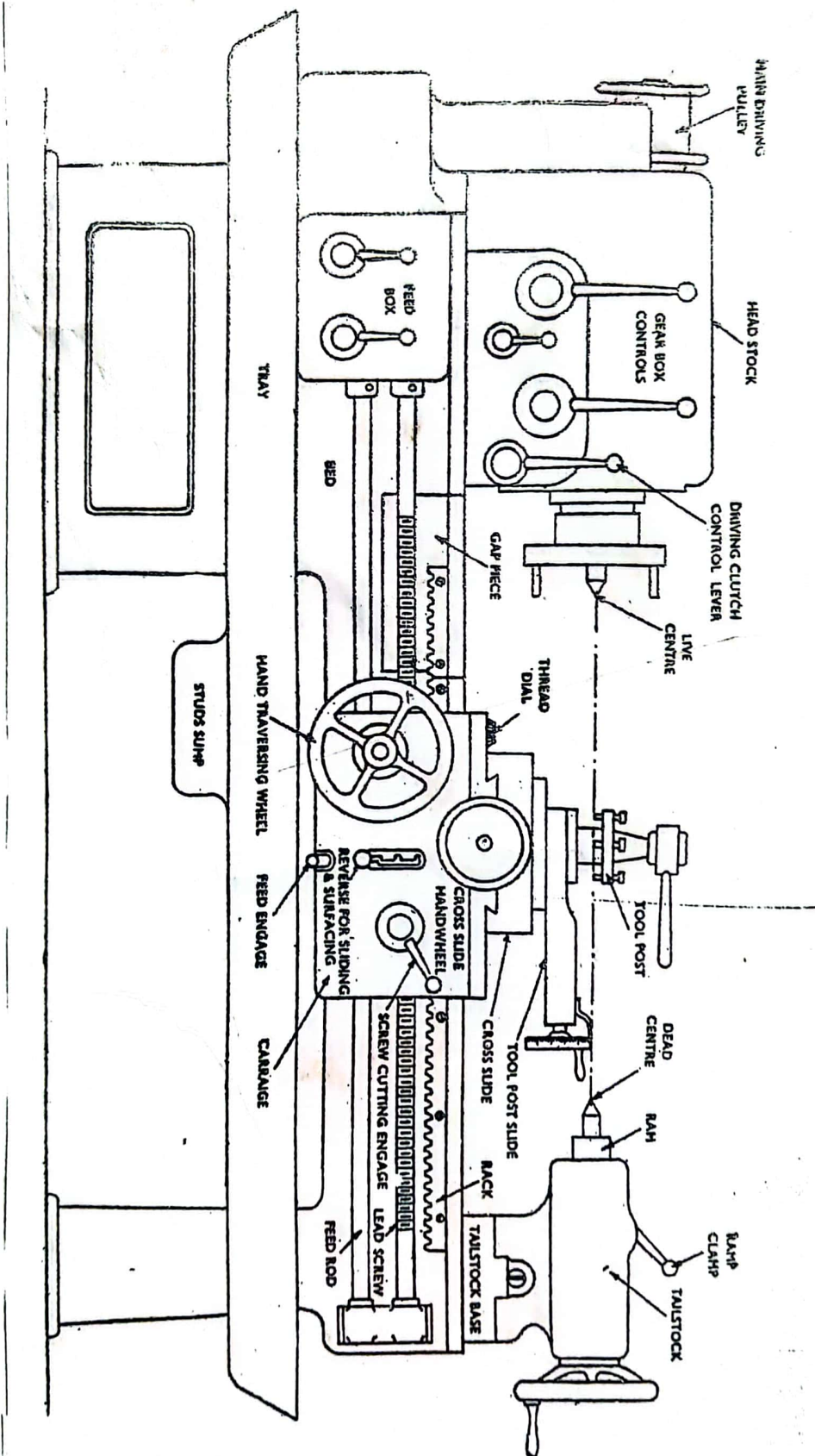




THE LATHE

Shop Safety

- The first step in preventing personal injury or machine damage in the shop is to make sure you are familiar with and know how to operate correctly the equipment you are using. If you don't know ask the area technician or shop personnel.
- ✓ ● Accidents are caused by inattention, taking chances, horseplay, bad judgment, fatigue, uncooperativeness, improper clothing, defective tools, etc.
- By following the safety rules given in the following pages and by cooperating with, and following any additional instructions the area technician or shop personnel might have.
- Please read over this manual carefully and follow the rules described. If you have any questions about the operation of any machine or tool, ask the area technician or personnel for instructions.

General Shop Safety Rules

- ✓ ● Safety glasses, and appropriate clothing and shoes are required at all times in the shop.
- Shoes must be worn in any shop area. The minimum footwear must cover the entire foot. No one wearing sandals will be allowed to use the shop space.
- Do not operate any equipment unless you are familiar with its operation. If you have any questions regarding the use of equipment ask the area personnel.
- In case of injury, report it to the shop personnel.
- If chemicals get in the eye, wash eye for 15 minutes in the eye wash.
- ✓ ● Do not wear ties, loose clothing, jewelry, gloves, etc. around moving or rotating machinery. Long hair must be tied back or covered to keep it away from moving machinery.
- ✓ ● Do not work in the shop if tired.
- ✓ ● Never indulge in horseplay in the shop areas.
- ✓ ● All machines must be operated with all required guards and shields in place.
- ✓ ● A brush, hood or special tool is preferred for removal of chips, shavings, etc. from the work area. Never use hands.

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- Keep fingers clear of the point of operation of machines by using special tools or devices, such as push sticks, hooks or pliers, etc. Never use a rag near moving machinery.
- A hard hammer should not be used to strike a hardened tool or any machine part. Use a soft faced hammer.
- Practice cleanliness in shop areas.
- Keep the floor around the machines clean, dry and free from trip hazards. Do not allow chips to accumulate.
- Before starting a machine, always check it for correct setup and always check to see if machine is clear by operating it by hand, if possible.
- Do not drink alcoholic beverages before or during work in the machine shop area. Do not bring food/snacks into the shop.
- Don't rush or take chances. Obey all safety rules.
- Heavy sanding and painting should only be done in well ventilated areas. preferably on the loading dock
- Follow all appropriate precautions when working with solvents, paints, adhesives or other chemicals. Use appropriate protective equipment.

Lathe Safety Rules

1. Move the tool bit a safe distance from the collet or chuck when inserting or removing work.
2. Don't run the machine faster than the proper cutting speed.
3. If any filing is done on work revolving in the lathe, file left handed to prevent slipping into the chuck. Never use a file without a handle.
4. Do not grasp or touch chips or turnings with your fingers, but get rid of them using a blunt instrument. It is safer to turn off the lathe before clearing chips then to leave it running.
5. Remove chuck key from chuck immediately after using
6. Turn chuck through by hand before turning on the power to be sure there is no binding or clearance problem.
7. Wear appropriate safety glasses. It may be necessary for others in the area to wear safety glasses too as objects will fly off the work.
8. Keep the floor free from obstructions, or slip hazards.
9. Ensure that lathe has a start/stop button within easy reach of the operator.
10. Follow job specifications for the speed, feed and depth of cut for materials being turned. Make sure all work runs true and centered.
11. Centre-drill work deeply enough to provide support for the piece while it is turning.

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12. Secure and clamp the piece being worked.
13. Adjust tool and tool rest so that they are slightly above the centre of the work.
14. Use a lifting device to handle heavy chucks or work.
15. Inspect chucks for wear or damage. Flying pieces can be very dangerous.
16. Use a barrier guard when operating the lathe in semi-automatic or automatic mode.
- ✓ 17. Guard all power transmission parts.
18. Remove all tools, measuring instruments and other objects from saddle or lathe bed before starting machine.
19. Keep all lathe cutting tools sharp.
- ✓ 20. Ensure that the chip and coolant shields are in place.
- ✓ 21. Shut off the power supply to the motor before mounting or removing accessories.
- ✓ 22. Stop lathe before taking measurements of any kind.
- ✓ 23. Use a vacuum, brush or rake to remove cuttings only after the lathe has stopped moving.
- ✓ 24. Keep working surface clean of scraps, tools and materials.
- ✓ 25. Keep floor around lathe clean and free of oil and grease.

Milling machine Safety Rules

1. Work must be clamped securely in a vise and vise clamped tightly to the table, or work must be clamped securely to the table.
2. Make sure cutter is rotating in the proper direction before cutting material.
- ✓ 3. Before running machine the spindle should be rotated by hand to make sure it is clear for cutting.
- ✓ 4. Make sure the power is off before changing cutters.
- ✓ 5. Always use the proper cutting fluid for the material being cut.
6. Never run the machine faster than the correct cutting speed.
7. Make sure that the machine is fully stopped before taking any measurements.
8. Always use cutters which are sharp and in good condition.
9. Don't place anything on the milling machine table such as wrenches, hammers, or tools.
- ✓ 10. Always stay at the machine while it is running.
11. Don't take too heavy a cut or use too rapid a feed.
- ✓ 12. Remove the collet tightening wrench immediately after using it.
- ✓ 13. Before cleaning the mill remove cutting tools from the spindle to avoid cutting yourself.

Shaping Machine Safety Rules

1. Do not use this machine unless a teacher has instructed you in its safe use and operation and has given permission.
2. Ensure no slip/trip hazards are present in workspaces and walkways.
3. Check that machine guards are in position.
4. Do not leave equipment on top of the machine.
5. Ensure cutter is in good condition and securely mounted.
6. Locate and ensure you are familiar with the operation of the ON/OFF starter
7. Faulty equipment must not be used. Immediately report suspect machinery.
8. Keep clear of moving machine parts.
9. Never leave the machine running unattended.
10. Do not leave equipment on top of the machine.
11. Follow correct clamping procedures- keep overhangs as small as possible and check workpiece is secure.
12. Ensure you have selected correct speed and rate.
13. Ensure you have set the correct depth of cut.
14. Before making adjustments and measurements or before cleaning swarf accumulations switch off and bring the machine to a complete standstill.
15. Switch off the machine.
16. Leave the machine and work area in a safe, clean and tidy state.

Drilling Safety Rules

1. Run drill at correct RPM for material, and drill bit size. General rule is the harder the material or the bigger the drill bit size, the slower the RPM.
2. Always hold work in a vise or clamp to the drill table.
3. Use the proper cutting fluid for the material being drilled. Ask the shop personnel about the appropriate fluid for the material you are machining.
4. Remove chips with a brush, never by hand.
5. Ease up on drilling pressure as the drill starts to break through the bottom of the material.
6. Don't drill with too much pressure.
7. Always try to support part on parallels or a backing board when drilling through material.
8. Always clean drill shank, sleeve and spindle hole before mounting bit.

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9. Never try to loosen the drill chuck while the power is on.
10. Never clean the machine while it is still in motion.
11. If the drill binds in a hole, stop the machine and turn the spindle backwards by hand to release the bit.
12. When drilling a deep hole, withdraw the bit frequently to clear chips.
13. Always remove the drill chuck key from the spindle immediately after using.
- ✓ 14. Wear safety eye protection while drilling.
15. Let the spindle stop of its own accord after turning the power off. Never try to stop the spindle with your hand.
16. Plexiglas and other brittle plastics can be difficult to drill. Ask the shop personnel for advice on drill and coolant selection when drilling these materials

Power Hacksaw Safety Rules

1. Ensure no slip/trip hazards are present in workspaces and walkways.
2. Locate and ensure you are familiar with the operation of the ON/OFF starter.
3. Ensure the material is tightly clamped in the work vice.
4. Check coolant delivery system to allow for sufficient flow of coolant.
5. Faulty equipment must not be used. Report suspect machinery immediately.
6. Ensure that the workpiece is secure.
7. Support overhanging work and signpost if it presents a hazard.
8. Ensure no-stands in front of the saw when it is started.
9. Keep clear of moving machine parts.
10. Direct coolant onto blade before starting cut.
11. Never leave the machine running unattended.
12. Before making adjustments or before cleaning swarf accumulations switch off and bring the machine to a complete standstill
13. Switch off machine.
14. Leave the machine and work area in a safe, clean and tidy state.

Grinding Machine Safety Rules

1. Wear goggles for all grinding machine operations.
2. Check grinding wheels for cracks before mounting.
3. Never operate grinding wheels at speeds in excess of the recommended speed.
4. Never adjust the workpiece or work mounting devices when the machine is operating
5. Do not exceed recommended depth of cut for the grinding wheel or machine

6. Remove workpiece from grinding wheel before turning machine off.
7. Use proper wheel guards on all grinding machines. On bench grinders, adjust tool rest 1/16 to 1/8 inch from the wheel.

Slotting Machine Safety Rules

1. Do not operate machines unless authorised and fully instructed on their operation and manner of action.
2. The machines may be only used for the functions they are intended for and in compliance with the received instructions.
3. Before starting your machine check whether it is properly functioning and all safeguards are in place.
4. Safeguards and protective devices should not be bypassed, removed or made nonfunctional.
5. Shut off the machine when not in use.
6. In case of failure never switch on any machine functions by forcedly activating the available limit switches.
7. Do not remove any cut cuttings manually, use the special tools (aids) provided for this purpose. For all operations required on the machine use only suitable hand tools.
8. Keep the tools, cleaning rags and machine attachments only in places intended for the purpose, not in the machine body, the gear box, etc.
9. Good order and cleanliness are the two essential prerequisites for secure operation. Therefore, keep the floor free and clean from any grease, oil and cut cuttings.
10. Place workpieces and tools so as that they do not create risk for you and the other people present.
11. When using cooling fluid or oil protect appropriately your hands with protection equipment.
12. Coolant may not be used for washing hands.
13. Report immediately any damage to your managers.
14. Unauthorised repairs are unacceptable.
15. Use protective goggles if during work cut cuttings or dust are likely to be present around your face.
16. If there are specific instructions regarding the operation of specific types of machines, follow them closely.